

FB-Mamba, an Efficient Bidirectional State Space Model for Wearable Sensor-Based Gait Analysis

Peng Wu

Shandong University

Tianyou Li

Hangkai Liu

Haoran Zhong

Ziyun Ding

Yibin Li

Rui Song

Huanghe Zhang

zhanghuanghe@sdu.edu.cn

Research Article

Keywords: Gait analysis, state space models, Mamba, wearable sensors, deep learning

Posted Date: February 24th, 2026

DOI: <https://doi.org/10.21203/rs.3.rs-8933284/v1>

License:   This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: The authors declare no competing interests.

FB-Mamba, an Efficient Bidirectional State Space Model for Wearable Sensor-Based Gait Analysis

Peng Wu¹, Tianyou Li¹, Hangkai Liu¹, Haoran Zhong¹, Ziyun Ding², Yibin Li¹, Rui Song¹, and Huanghe Zhang³

February 21, 2026

Abstract

Wearable sensor-based gait analysis encompasses diverse tasks including human activity recognition, neurodegenerative disease diagnosis, and clinical event detection. Existing deep learning approaches are typically designed for specific tasks, lacking a unified framework that generalizes across different scenarios. We present the first systematic investigation of state space models for wearable sensor-based gait analysis, proposing FB-MAMBA, a Forward-Backward Mamba architecture that achieves bidirectional temporal modeling with linear complexity through an adaptive fusion mechanism. We establish FB-MAMBA as a general-purpose framework spanning three major task categories, including human activity recognition, disease detection, and event-level detection. Comprehensive experiments on five datasets with heterogeneous sensor modalities and sampling rates demonstrate that FB-MAMBA achieves consistently competitive performance across all task categories while maintaining computational efficiency and favorable long-range dependency modeling capability.

Keywords: Gait analysis, state space models, Mamba, wearable sensors, deep learning

1 Introduction

Gait analysis using wearable sensors has become increasingly important for clinical applications, including the detection of movement disorders [1], fall risk assessment [2–4], and rehabilitation monitoring [5]. Modern wearable sensing technologies encompass diverse modalities such as inertial measurement units (IMUs) combining accelerometers and gyroscopes for motion capture, smartphone-embedded sensors enabling ubiquitous activity monitoring, and force-sensitive footswitches measuring plantar pressure patterns for gait timing analysis [6, 7]. The continuous monitoring capability of these wearable sensors enables early detection of gait abnormalities and provides objective measures for treatment evaluation [8, 9]. Beyond general activity monitoring,

*This work was supported in part by the Young Scientists Fund of the National Natural Science Foundation of China under Grant 62403281, in part by the Taishan Scholars Project (Young Expert Program) under Grant NO.tsqn202408040, in part by the Shandong Excellent Young Scientists Fund Program (Overseas) under Grant 2024HWYQ-019, and in part by the Open Project Fund of International Joint Research Center for Perception and Control of Intelligent Rehabilitation Systems of Sichuan Province under Grant No.25-H-01. (Corresponding author: Huanghe Zhang)

^{†1}Peng Wu, Tianyou Li, Hangkai Liu, Haoran Zhong, Yibin Li, Rui Song and Huanghe Zhang (e-mail: zhanghuanghe@sdu.edu.cn) are with the Center for Robotics, School of Control Science and Engineering, Shandong University, Jinan, Shandong 250100, China

^{†2}Ziyun Ding (e-mail: z.ding@bham.ac.uk) is with the School of Engineering, University of Birmingham, Birmingham B15 2TT, United Kingdom.

^{§3}Huanghe Zhang is with the International Joint Research Center for Perception and Control of Intelligent Rehabilitation Systems of Sichuan Province, Sichuan, China.

specific clinical challenges include freezing of gait (FoG) detection in Parkinson’s disease [10,11], fall risk assessment in the elderly, and differential diagnosis of neurodegenerative conditions based on gait biomarkers.

The diverse requirements of wearable sensor-based gait analysis span three major task categories, namely human activity recognition for daily living monitoring, disease detection for clinical diagnosis, and event-level detection for real-time clinical intervention. Each category presents distinct technical challenges ranging from multi-class activity discrimination to binary disease classification and temporal event detection. Despite this diversity, most existing deep learning approaches are designed for individual tasks with task-specific architectures and pre-processing pipelines, lacking a unified framework that generalizes across different application scenarios. Addressing these heterogeneous requirements within a single model demands an architecture that can efficiently capture both short- and long-range temporal dependencies across diverse sensor modalities and clinical contexts.

Deep learning approaches have shown remarkable success in analyzing IMU signals for gait-related tasks [12–15]. Recurrent neural networks (RNNs), particularly Long Short-Term Memory (LSTM) networks [16], have been widely adopted due to their ability to model temporal dependencies. Bidirectional LSTMs (BiLSTMs) [17,18] further improve performance by capturing both past and future context. However, these models suffer from several fundamental limitations. First, the sequential nature of RNN computation prevents parallelization, resulting in slow training and inference. Second, despite the gating mechanisms, LSTMs still struggle with very long-range dependencies due to gradient flow issues [19]. Third, the hidden state bottleneck limits the model’s capacity to maintain rich representations over extended sequences.

Transformer architectures [20] address the long-range dependency problem through self-attention mechanisms that enable direct connections between any two positions in a sequence, but introduce quadratic computational complexity with respect to sequence length. This makes them impractical for processing long continuous sensor streams, particularly in continuous gait monitoring applications where real-time processing is essential [21,22]. Although various efficient Transformer variants have been proposed, such as sparse attention patterns [23], these approaches still face challenges on resource-constrained wearable devices where computational resources and battery life are limited.

Recently, state space models (SSMs), particularly Mamba [24], have emerged as an efficient alternative that achieves linear complexity while maintaining strong modeling capabilities. Building upon the foundational work of Structured State Space Models (S4) [25] and its variants [26,27], Mamba introduces selective state spaces with input-dependent parameters that adapt to the input sequence. Extensions to vision tasks [28,29] and language modeling [30–32] have demonstrated the versatility of SSM-based architectures across diverse domains.

However, the original Mamba architecture processes sequences unidirectionally, which may not fully capture the bidirectional nature of gait patterns. In gait analysis, both preceding and following movements provide important context for understanding the current state. For example, distinguishing between similar activities (e.g., walking vs. walking upstairs) benefits from observing both the initial movement patterns and subsequent motion characteristics. Similarly, detecting clinical events like falls or gait abnormalities benefits from bidirectional temporal context. This observation motivates the development of bidirectional SSM architectures that can efficiently capture context from both past and future while maintaining the computational advantages of linear complexity.

In this paper, we propose FB-MAMBA, a novel architecture that combines the efficiency of state space models with bidirectional temporal modeling. Unlike simple concatenation approaches used in BiLSTMs, our method employs an adaptive fusion mechanism that weights the contributions from forward and backward processing based on the input context. This design enables the model to dynamically determine the relative importance of past and future information for each prediction, leading to more effective temporal modeling. Our key contributions

include the following four aspects.

1) We present the first systematic study of state space models for wearable sensor-based gait analysis. We introduce FB-MAMBA and evaluate it across three representative task categories, namely human activity recognition, disease detection, and event-level detection.

2) We propose a unified framework that handles multi-class activity classification, binary disease diagnosis, and clinical event detection within a single architecture, without task-specific modifications.

3) We design an efficient bidirectional architecture with linear complexity. The Forward-Backward Mamba processes sequences in both directions using parallel state space models with an adaptive fusion mechanism that learns input-dependent weights for combining temporal directions. This achieves bidirectional modeling capability while maintaining $O(L)$ complexity.

4) We validate FB-MAMBA across five datasets (Daphnet, UCI-HAR, Huntington, ALS, and MobiFall) spanning different sensor modalities (IMU, smartphone, footswitch), sampling rates (50–300 Hz), and clinical applications, demonstrating consistently competitive performance across all task categories.

2 Related Work

2.1 Deep Learning for Gait Analysis

Deep learning has revolutionized sensor-based gait analysis. Ordóñez and Roggen [33] proposed DeepConvLSTM, combining convolutional layers with LSTM for activity recognition. Hammerla *et al.* [34] conducted a comprehensive comparison of deep learning architectures for wearable activity recognition. Temporal Convolutional Networks (TCN) [35] have shown competitive performance with recurrent models while enabling parallel computation. More recently, InceptionTime [36] and ROCKET [37] have achieved strong results on time series classification benchmarks.

These methods have been applied across various gait analysis task categories. For clinical applications such as freezing of gait detection, Camps *et al.* [38] developed deep learning systems using waist-worn IMUs, while Sigcha *et al.* [39] compared various architectures for FoG detection. For fall detection and disease diagnosis, machine learning methods have shown promising results on gait abnormality classification [40]. Recent studies have also demonstrated that the accuracy of wearable-derived gait parameters significantly impacts fall risk prediction performance [41].

Transformer-based models have also been applied to gait analysis and human activity recognition [42], achieving strong performance but with high computational costs that limit deployment on wearable devices. Recent advances in deep learning for gait analysis include two-dimensional deep convolutional networks for stride parameter estimation [43], transductive learning approaches for ambulatory gait analysis [44], and deep neural networks for gait cycle percentage prediction [45].

2.2 State Space Models

Structured State Space Models (S4) [25] introduced an efficient approach for modeling long sequences by leveraging the structure of state space representations. The S4D variant [26] simplified the parameterization while maintaining performance. Subsequent works have explored various improvements, including gating mechanisms in H3 [30], implicit convolutions in Hyena [31], and a combination of RNN and Transformer concepts in RWKV [32].

Mamba [24] further improved upon S4 by introducing selective state spaces with input-dependent parameters, achieving competitive performance with Transformers while maintaining linear complexity. Recent work has established theoretical connections between Transformers

and SSMs [46]. Vision Mamba [28] and VMamba [29] have extended these ideas to visual recognition tasks, demonstrating the broad applicability of SSM-based architectures.

However, existing SSM-based approaches primarily focus on unidirectional processing, which may limit their effectiveness for tasks requiring bidirectional context understanding.

2.3 Bidirectional Sequence Modeling

Bidirectional processing has been shown to improve performance in various sequence modeling tasks [18]. BiLSTMs [17] process sequences in both directions and concatenate the outputs. BERT [47] demonstrated the power of bidirectional pre-training for language understanding. However, these approaches either double the computational cost (BiLSTM) or require quadratic complexity (BERT), leaving a gap for efficient bidirectional architectures with linear scaling.

For gait analysis, bidirectional modeling is particularly important because gait patterns exhibit temporal symmetry, and context from both past and future movements can improve detection accuracy [48, 49].

3 Method

3.1 Background on State Space Models

State space models define a linear mapping from input $x(t) \in \mathbb{R}$ to output $y(t) \in \mathbb{R}$ through a hidden state $h(t) \in \mathbb{R}^N$.

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \quad (1)$$

$$y(t) = \mathbf{C}h(t) + \mathbf{D}x(t) \quad (2)$$

where $\mathbf{A} \in \mathbb{R}^{N \times N}$, $\mathbf{B} \in \mathbb{R}^{N \times 1}$, $\mathbf{C} \in \mathbb{R}^{1 \times N}$, and $\mathbf{D} \in \mathbb{R}$.

For discrete sequences, the continuous system is discretized using a step size Δ .

$$\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}) \quad (3)$$

$$\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B} \quad (4)$$

Mamba [24] introduces selective state spaces where parameters $\mathbf{B}$, $\mathbf{C}$, and Δ are input-dependent. This enables adaptive processing that responds to the input sequence.

$$\mathbf{B}_t = \text{Linear}_B(x_t) \quad (5)$$

$$\mathbf{C}_t = \text{Linear}_C(x_t) \quad (6)$$

$$\Delta_t = \text{softplus}(\text{Linear}_\Delta(x_t)) \quad (7)$$

3.2 FB-Mamba Architecture

Our proposed FB-MAMBA architecture consists of three main components, namely input projection, forward-backward Mamba blocks, and task-specific heads. The overall architecture is illustrated in Fig. 1.

3.2.1 Input Projection

Given an input sensor sequence $\mathbf{X} \in \mathbb{R}^{T \times D}$ where T is the sequence length (set to 128 in the main experiments) and $D \in \{2, 6, 9\}$ is the input dimension depending on the sensor configuration (2 for footswitch, 6 for phone/IMU, 9 for 3-IMU), we first project it to a higher-dimensional feature space.

$$\mathbf{H}_0 = \mathbf{X}\mathbf{W}_{\text{proj}} + \mathbf{b}_{\text{proj}} \quad (8)$$

where $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{D \times d}$, $\mathbf{b}_{\text{proj}} \in \mathbb{R}^d$, and d is the model dimension. The output $\mathbf{H}_0 \in \mathbb{R}^{T \times d}$ serves as the input to the FB-Mamba blocks.

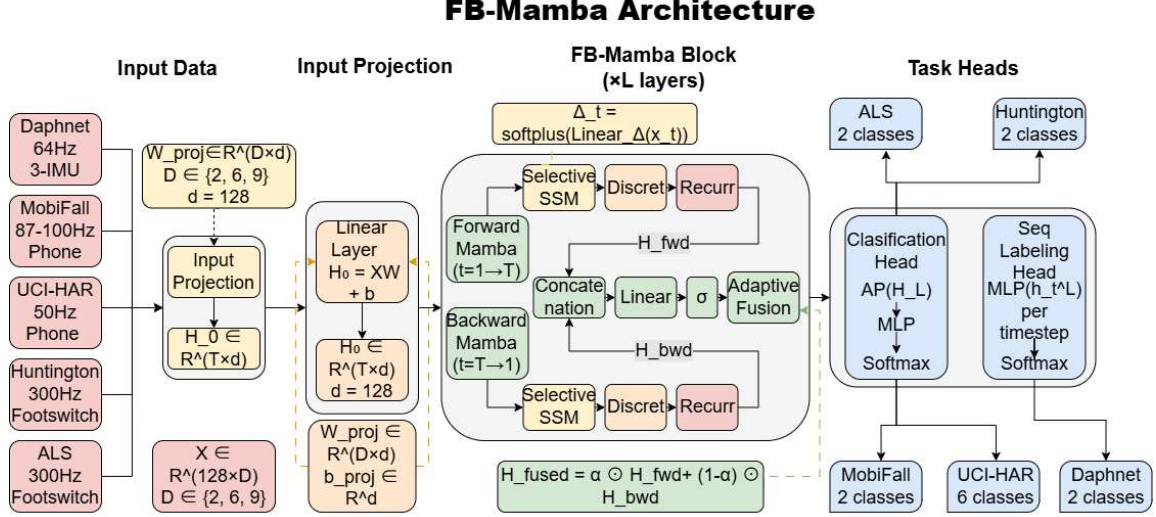


Figure 1: Overview of the FB-Mamba architecture. The input sensor data is first projected to a d -dimensional feature space. The FB-Mamba block processes the projected features bidirectionally through parallel forward and backward Selective SSM branches, whose outputs are combined via learned adaptive fusion weights α . The fused representations are passed to task-specific heads for classification or per-timestep sequence labeling. Detailed formulations are provided in Section 3.

3.2.2 Forward-Backward Mamba Block

The core of our architecture is the Forward-Backward Mamba block, which is stacked across multiple layers. This block processes sequences bidirectionally. For input $\mathbf{H} \in \mathbb{R}^{T \times d}$, the processing consists of three steps.

Forward Pass. The forward Mamba processes the sequence from $t = 1$ to $t = T$, employing the Selective SSM with discretization and recurrence operations.

$$\mathbf{H}_{\text{fwd}} = \text{Mamba}_{\text{fwd}}(\mathbf{H}) \quad (9)$$

Each Mamba module contains a Selective SSM layer that computes input-dependent parameters for discretization and applies recurrent state updates.

Backward Pass. The backward Mamba processes the sequence from $t = T$ to $t = 1$ by first reversing the input, applying the Selective SSM, and then reversing the output back to the original order.

$$\mathbf{H}_{\text{bwd}} = \text{flip}(\text{Mamba}_{\text{bwd}}(\text{flip}(\mathbf{H}))) \quad (10)$$

Adaptive Fusion. We employ an adaptive fusion mechanism that combines forward and backward representations, specifically designed for gait analysis tasks. Unlike simple concatenation used in BiLSTMs, our approach is motivated by two key observations. First, gait signals exhibit temporal asymmetry where the importance of past versus future context varies across different phases of the gait cycle. Second, clinical events and activity transitions require adaptive weighting of contextual information for accurate boundary detection. The forward and backward outputs are first concatenated, then passed through a linear layer followed by sigmoid activation to compute the fusion weights:

$$\alpha = \sigma(\mathbf{W}_\alpha [\mathbf{H}_{\text{fwd}}; \mathbf{H}_{\text{bwd}}] + \mathbf{b}_\alpha) \quad (11)$$

The fused representation is computed as a weighted combination:

$$\mathbf{H}_{\text{fused}} = \alpha \odot \mathbf{H}_{\text{fwd}} + (1 - \alpha) \odot \mathbf{H}_{\text{bwd}} \quad (12)$$

Finally, the output is processed through a linear layer with layer normalization:

$$\mathbf{H}_{\text{out}} = \text{LayerNorm}(\mathbf{W}_{\text{out}}\mathbf{H}_{\text{fused}} + \mathbf{b}_{\text{out}}) \quad (13)$$

where σ is the sigmoid function, $\odot$ denotes element-wise multiplication, and $[\cdot; \cdot]$ denotes concatenation.

3.2.3 Task Heads

For classification tasks, we apply global average pooling followed by a classification head.

$$\hat{y} = \text{softmax}(\text{MLP}(\text{AvgPool}(\mathbf{H}_{\text{out}}))) \quad (14)$$

For sequence labeling tasks such as FoG detection, we apply a per-timestep classifier.

$$\hat{y}_t = \text{softmax}(\text{MLP}(\mathbf{h}_{t,\text{out}})) \quad (15)$$

3.3 Complexity Analysis

The computational complexity of FB-MAMBA is $O(2L \cdot d \cdot N)$, where L is sequence length, d is model dimension, and N is state dimension. The factor of 2 accounts for the parallel forward and backward SSM passes. The parallel processing of forward and backward passes can be efficiently implemented on modern hardware, enabling FB-Mamba to achieve both bidirectional context modeling and computational efficiency. A detailed comparison with baseline methods is provided in Table 3 in Section 5.

4 Experiments

4.1 Datasets

We evaluate FB-MAMBA on five datasets with validated ground truth labels.

Daphnet provides FoG detection data from 10 Parkinson’s disease patients. The dataset includes 3-axis accelerometer data from sensors placed on the ankle, thigh, and trunk, sampled at 64 Hz. FoG episodes were annotated by movement disorder specialists with high inter-rater agreement ($\kappa > 0.85$) [10, 50].

UCI-HAR contains 6-class activity recognition data from 30 subjects. Data was collected at 50 Hz using embedded smartphone accelerometer and gyroscope. Activities were performed in controlled laboratory conditions with continuous supervision [51].

PhysioNet Huntington’s Disease provides gait data from 15 patients with Huntington’s disease and 16 healthy controls. The dataset includes stride-to-stride fluctuations measured using force-sensitive foot switches during 5-minute walking trials at 300 Hz. Disease diagnosis was confirmed by neurologists based on established clinical criteria [52, 53].

PhysioNet ALS contains gait data from 13 patients with Amyotrophic Lateral Sclerosis (ALS) and 16 healthy controls, following the same data collection protocol. ALS diagnosis was confirmed using the El Escorial criteria [52, 53].

MobiFall provides fall detection data from 9 subjects using a Samsung Galaxy S3 smartphone worn at the waist. The dataset includes 6 channels of inertial data (3-axis accelerometer and 3-axis gyroscope) sampled at approximately 87–100 Hz using the SENSOR_DELAY_FASTEST setting. We extract accelerometer and gyroscope readings from the raw files, concatenating them into 6-channel input. The binary classification task distinguishes between fall events and activities of daily living (ADL) [54].

Table 1 summarizes the dataset characteristics, and Fig. 2 illustrates the data acquisition setups for each dataset.

Table 1: Dataset characteristics. Input dimension D corresponds to the number of sensor channels.

Characteristic	Daphnet	UCI-HAR	Huntington	ALS	MobiFall
Sensor Type	3-IMU	Phone	Footswitch	Footswitch	Phone
Sampling Rate	64 Hz	50 Hz	300 Hz	300 Hz	87–100 Hz
Channels (D)	9	6	2	2	6
Classes	2	6	2	2	2
Subjects	10	30	31	29	9

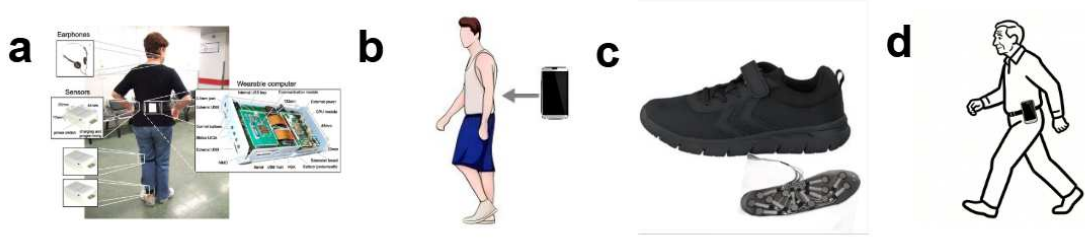


Figure 2: Data acquisition setups for the five datasets: (a) Daphnet [50], (b) UCI-HAR [51], (c) PhysioNet (Huntington/ALS) [52, 53], (d) MobiFall [54].

4.2 Data Preprocessing

All datasets are segmented using a sliding window approach with a fixed window size of 128 samples. Given the different sampling rates across datasets, this corresponds to 2.0 seconds for Daphnet (64 Hz), 2.56 seconds for UCI-HAR (50 Hz), approximately 1.3–1.5 seconds for MobiFall (87–100 Hz), and 0.43 seconds for PhysioNet datasets (300 Hz). We use 50% overlap between consecutive windows to increase the number of training samples and capture activity transitions more effectively.

For each dataset, we apply z-score normalization to each sensor channel independently. The normalization parameters (mean and standard deviation) are computed on the training set and applied to both training and test sets to prevent data leakage.

4.3 Baselines

We compare FB-MAMBA with representative methods spanning different architectural paradigms for time series analysis. DNN, CNN-LSTM, BiLSTM, and Mamba are evaluated across all datasets in the main accuracy comparison (Table 2), while TCN and Transformer are additionally included in the computational efficiency analysis (Table 3).

DNN is a 3-layer fully connected network with hidden dimensions of 256-128-64, using ReLU activation and dropout rate of 0.3 between layers. The input sensor windows are flattened before feeding into the network.

CNN-LSTM consists of 4 convolutional layers with 64-128-128-128 filters, kernel size of 5, followed by 2 LSTM layers with 128 hidden units each. Batch normalization and ReLU activation are applied after each convolutional layer [33].

BiLSTM employs 2 bidirectional LSTM layers with 128 hidden units per direction (256 total per layer). The outputs from both directions are concatenated, followed by a linear classification layer [17, 18].

TCN implements 4 residual blocks with dilation factors of 1, 2, 4, 8, kernel size of 3, and 128 channels per layer. Each block contains causal convolution, weight normalization, ReLU activation, and dropout (0.2) [35].

Transformer uses 4 encoder layers with 4 attention heads, model dimension of 128, and feedforward dimension of 256. Positional encoding is added to the input embeddings, followed by layer normalization [20].

Mamba consists of 3 Mamba blocks with model dimension $d = 256$ and state dimension $N = 16$. Each block contains selective SSM with input-dependent discretization parameters [24].

4.4 Implementation Details

All models are trained using the AdamW optimizer [55] with a learning rate of 10^{-3} and weight decay of 10^{-4} . We use a cosine annealing learning rate schedule and early stopping with patience of 15 epochs. Batch size is 32 and the maximum number of epochs is 100.

For FB-MAMBA, we use $d_{model} = 256$ (model dimension), $d_{state} = 16$ (state dimension), and 3 FB-Mamba layers. Sequence length is 128 samples. All experiments are conducted on NVIDIA A100 GPUs.

4.5 Evaluation Protocol

Due to the limited number of subjects in these datasets (10 patients in Daphnet, 30 subjects in UCI-HAR, 31 subjects in Huntington, 29 subjects in ALS, and 9 subjects in MobiFall), we adopt the following evaluation strategies to ensure robust performance estimation.

For Daphnet, Huntington, ALS, and MobiFall, we employ 10-fold cross-validation. The data is randomly partitioned into 10 folds, with 9 folds used for training and 1 fold for testing in each iteration. We report the mean and standard deviation across all folds.

For UCI-HAR, we follow the original dataset split provided by [51], where 21 subjects are used for training and 9 subjects for testing. This predefined split ensures comparability with published results.

5 Results

5.1 Comparison with Existing Models

Table 2 presents a comprehensive comparison with existing models across all five datasets. FB-MAMBA achieves competitive performance across all datasets while maintaining favorable parameter count. For Daphnet and UCI-HAR, we include published results from prior work for reference. For PhysioNet disease detection datasets (Huntington and ALS), published baseline results are limited as these represent relatively unexplored task formulations, and we therefore compare against our own implementations using identical experimental protocols. All our implemented baselines and FB-MAMBA use 10-fold cross-validation except UCI-HAR which follows the original train/test split. Published results from Prabhu *et al.* [56] and Setiawan *et al.* [57] on PhysioNet datasets use LOOCV.

Fig. 3 provides a visual comparison of the performance across all methods and datasets. FB-MAMBA achieves 95.29% accuracy on Daphnet (FoG detection), 98.83% on UCI-HAR (6-class activity recognition), 99.17% on Huntington’s disease detection, 99.56% on ALS detection, and 99.58% on MobiFall (fall detection), demonstrating consistent competitive performance across all task categories.

5.2 Computational Efficiency Analysis

Table 3 compares computational efficiency across methods, including parameter count, memory usage, throughput, and time complexity.

FB-MAMBA achieves $1.22\times$ higher throughput compared to Transformer (9,900 vs 8,100 samples/s) and $1.9\times$ higher throughput than BiLSTM while maintaining competitive memory

Table 2: Performance comparison across five datasets. Superscripts on published results denote differences in evaluation protocol relative to our setup. [‡]LOOCV. [§]Fixed split. ^{||}Three-class FoG. [#]70/30 split. [△]22/8 subject split. ^{*}10-fold CV. [◇]Instance-level validation. [♣]Merged MobiFall+MobiAct. [♠]Merged MobiFall+SisFall (8-class). [♡]Three-class FoG prediction.

Method	Daphnet		UCI-HAR		Huntington		ALS		MobiFall	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
<i>Published Results</i>										
Hammerla [§] [34]	—	76.0	—	—	—	—	—	—	—	—
Kleanthous [58]	79.9	—	—	—	—	—	—	—	—	—
LSTM-CNN [△] [59]	—	—	95.8	95.8	—	—	—	—	—	—
Prabhu [‡] [56]	—	—	—	—	100	—	96	—	—	—
CNN-WCS [‡] [57]	—	—	—	—	91.67	—	96.55	—	—	—
Erdaş [*] [60]	—	—	—	—	84.21	84.36	86.12	82.11	—	—
Andrei [*] [61]	90.92	—	—	—	—	—	—	—	—	—
Wan [#] [62]	—	—	92.71	92.93	—	—	—	—	—	—
HACNN [◇] [63]	—	—	—	—	—	—	—	—	—	98.08
TriNet [♣] [64]	—	—	—	—	—	—	—	—	95.92	—
FD-CNN [♠] [65]	—	—	—	—	—	—	—	—	98.61	—
FoG-M3 [♡] [66]	95.3	—	—	—	—	—	—	—	—	—
<i>Baselines (Our Implementation)</i>										
DNN	93.74	54.47	89.81	90.51	96.80	96.78	97.50	97.48	97.23	94.85
CNN-LSTM [33]	92.69	55.56	95.73	96.06	97.50	97.48	98.20	98.18	98.42	96.18
BiLSTM [17]	93.23	56.04	95.53	95.89	97.65	97.63	98.35	98.33	98.76	96.89
Mamba [24]	94.11	61.70	98.64	98.74	99.17	99.16	99.51	99.50	99.35	96.52
FB-Mamba	95.29	63.45	98.83	98.93	99.17	99.16	99.56	99.51	99.58	97.78

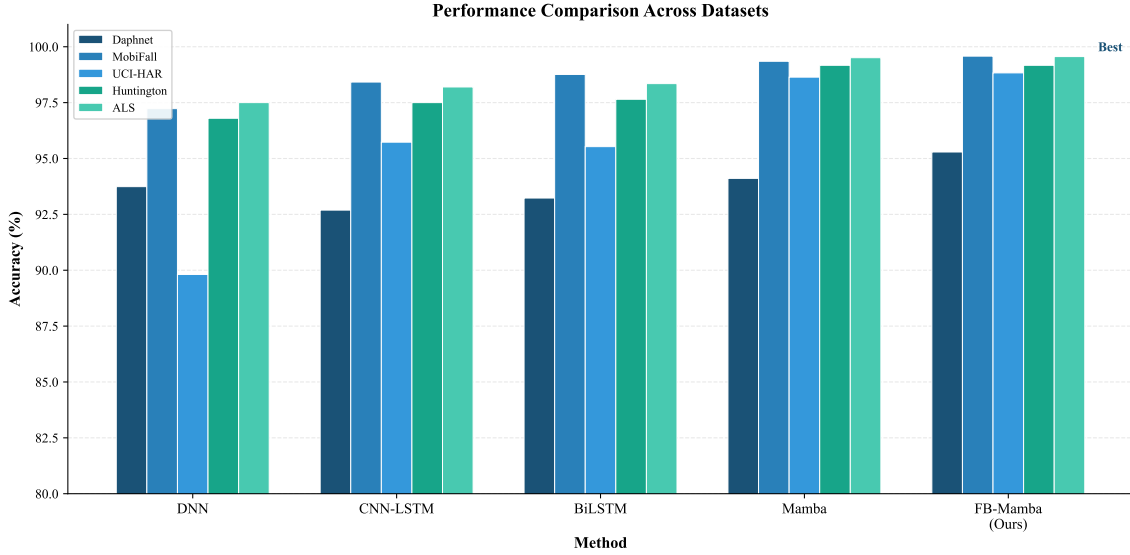


Figure 3: Performance comparison across all methods and datasets using grouped bar chart visualization.

usage. Fig. 4 provides a comprehensive visualization of the efficiency trade-offs across multiple dimensions, demonstrating that FB-MAMBA achieves the best balance between accuracy and computational cost.

Table 3: Computational efficiency comparison. Throughput measured with batch size 32, sequence length 128. L : sequence length, D : input dimension, d : model dimension, h : hidden dimension, N : state dimension, K : kernel size.

Model	Params (K)	Memory (MB)	Throughput (samples/s)	Time Complexity
DNN	850	120	12,500	$O(D \cdot L)$
CNN-LSTM	1,800	210	4,800	$O(K \cdot L + L \cdot h^2)$
BiLSTM	1,200	180	5,200	$O(2L \cdot h^2)$
TCN	2,100	250	6,800	$O(K \cdot d \cdot L \log L)$
Transformer	1,500	420	8,100	$O(L^2 \cdot d)$
Mamba	1,100	320	9,200	$O(L \cdot d \cdot N)$
FB-Mamba	2,900	490	9,900	$O(2L \cdot d \cdot N)$

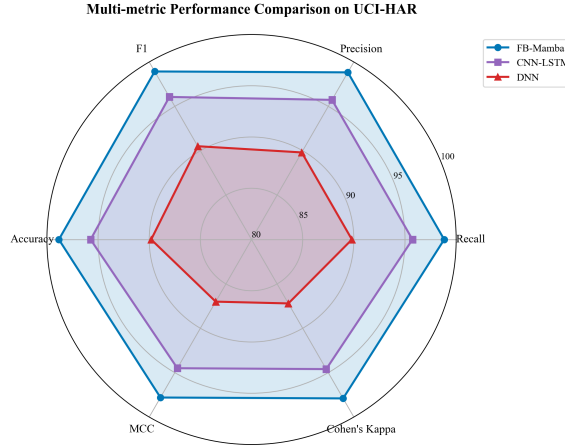


Figure 4: Multi-dimensional efficiency comparison using radar chart visualization across five dimensions: Parameters, Inference Time, Memory Usage, Throughput, and Accuracy.

5.3 Ablation Study

To analyze the contribution of each architectural component, we conduct ablation studies on the ALS dataset. Table 4 presents the results.

The ablation results on the ALS dataset provide preliminary insights into component contributions: (1) Bidirectional processing appears beneficial, with forward-only showing -2.98% and backward-only showing -3.14% compared to the full model. (2) The adaptive fusion mechanism shows +3.21% accuracy over simple concatenation, suggesting that learned input-dependent weighting may be useful for combining bidirectional representations. (3) Using separate parameters for forward and backward branches seems important, as the shared-weights variant shows -3.35% accuracy drop.

5.4 Sensitivity Analysis

Table 5 presents the effect of model configuration on performance. We evaluate the impact of state dimension d_{state} , number of layers n_{layers} , and model dimension d_{model} on the UCI-HAR dataset. The default configuration ($d_{model}=256$, $d_{state}=16$, $n_{layers}=3$) provides a reasonable

Table 4: Ablation study on ALS dataset. Results show the contribution of each component to FB-Mamba’s performance.

Variant	Accuracy (%)	Std (%)	ΔAcc
Full FB-Mamba	99.56	± 0.28	–
Forward-Only	96.58	± 0.89	-2.98
Backward-Only	96.42	± 0.95	-3.14
Shared-Weights	96.21	± 1.02	-3.35
Concat-Fusion	96.35	± 0.88	-3.21

Table 5: Model configuration analysis on UCI-HAR. Default configuration: $d_{\text{model}}=256$, $d_{\text{state}}=16$, $n_{\text{layers}}=3$.

Parameter	Value	Acc (%)	F1 (%)
d_{state}	8	98.24	98.35
	16	98.83	98.93
	32	98.80	98.89
	64	98.61	98.69
n_{layers}	1	97.94	98.04
	2	98.57	98.66
	3	98.83	98.93
	4	98.77	98.86
d_{model}	64	96.87	96.96
	128	98.17	98.26
	256	98.83	98.93
	512	98.90	98.99

balance between accuracy and efficiency, achieving 98.83% accuracy. Increasing model capacity beyond this point yields diminishing returns, with only +0.07% accuracy gain from $d_{\text{model}}=256$ to 512 while substantially increasing computational cost.

5.5 Scalability Analysis

The inference time scaling with sequence length demonstrates that FB-MAMBA maintains linear complexity, while Transformer exhibits quadratic growth. Fig. 5 shows the time complexity comparison across different sequence lengths.

At sequence length 1024, FB-MAMBA achieves $12.7\times$ speedup over Transformer, enabling real-time processing of extended gait sequences on resource-constrained devices [67, 68]. The practical benefits of linear complexity become increasingly pronounced at longer sequences, making FB-MAMBA well suited for continuous gait monitoring applications where extended time windows may be necessary.

5.6 Long Sequence Performance Analysis

To evaluate long-range dependency modeling beyond the 128-sample windows used in the main experiments, we design a FoG event counting task on the Daphnet dataset. Unlike per-timestep FoG detection, this task requires the model to integrate information across the entire sequence to determine the total number of FoG episodes. We construct long sequences using a sliding window approach. For a given sequence length L , we sample non-overlapping windows from continuous recordings and concatenate them to form extended sequences. The ground truth

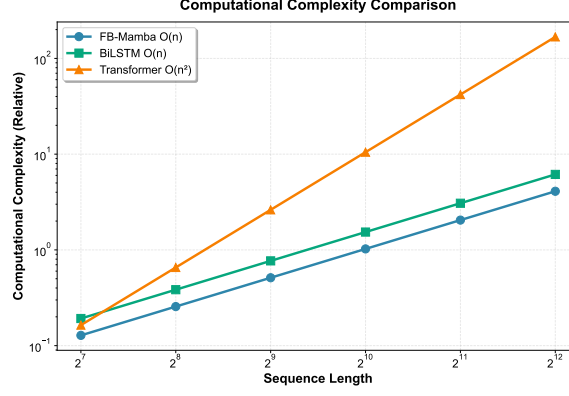


Figure 5: Time complexity comparison across different sequence lengths.

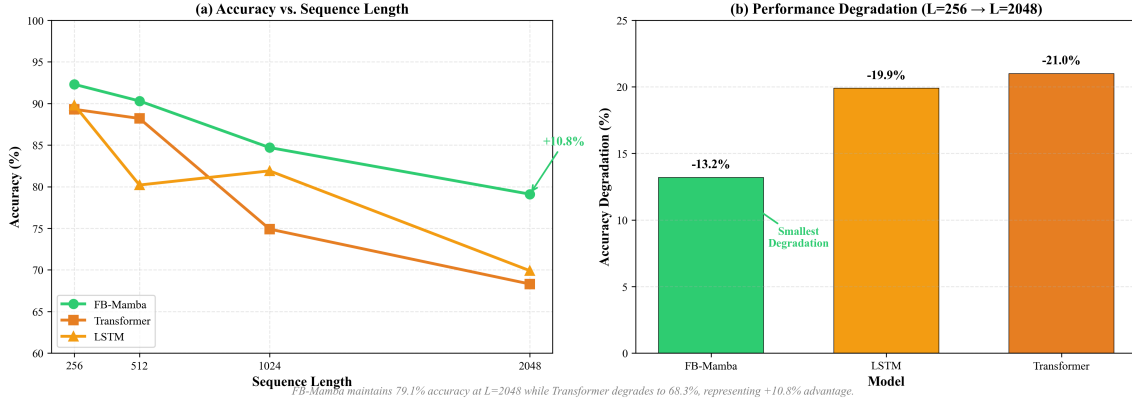


Figure 6: Long sequence performance comparison showing FoG event counting accuracy across different sequence lengths ($L=256, 512, 1024, 2048$).

event count N_{FoG} is computed as:

$$N_{\text{FoG}} = \sum_{i=1}^L \mathbf{1}[y_i = 1 \wedge y_{i-1} = 0] \quad (16)$$

where $y_i \in \{0, 1\}$ indicates FoG presence at timestep i , and the indicator function counts FoG onset events (transitions from non-FoG to FoG state).

We report ExactAcc (exact match accuracy), which requires the model to predict the precise number of FoG events. This strict metric evaluates whether the model can reliably count discrete events across long temporal spans, relevant for clinical applications such as assessing disease severity based on FoG episode frequency.

As shown in Table 6, FB-Mamba exhibits the smallest performance degradation (-13.2%) as sequence length increases from 256 to 2048, compared to Transformer (-21.0%) and LSTM (-19.9%). At $L=2048$, FB-Mamba achieves 79.1% accuracy versus 68.3% for Transformer (+10.8%), suggesting its suitability for continuous gait monitoring applications where extended observation windows are required.

6 Discussion

We organize our discussion around the three task categories examined in this study, analyzing how the architectural design of FB-Mamba addresses the specific requirements of each application domain.

Table 6: Long sequence FoG event counting performance on Daphnet dataset. ExactAcc measures exact match between predicted and actual FoG event counts. Degradation indicates accuracy drop from L=256 to L=2048.

Model	L=256	L=512	L=1024	L=2048	Degrad.
FB-Mamba	92.3%	90.3%	84.7%	79.1%	-13.2%
Transformer	89.3%	88.2%	74.9%	68.3%	-21.0%
LSTM	89.8%	81.9%	80.2%	69.9%	-19.9%

For human activity recognition, FB-Mamba achieves the highest accuracy among all compared methods on the UCI-HAR dataset for 6-class activity classification. The bidirectional processing is beneficial for distinguishing activities that share similar instantaneous sensor patterns but differ in temporal context, such as walking versus walking upstairs. The selective state space mechanism adaptively adjusts its parameters at each timestep, and the bidirectional extension allows the adaptive fusion weights to reflect the relative importance of preceding and subsequent context for each activity class. The ablation study confirms that both temporal directions contribute to the classification decision, consistent with the observation that human activities exhibit characteristic signatures in both their onset and completion phases.

For disease detection, the Huntington and ALS datasets represent neurodegenerative disease diagnosis based on gait abnormality patterns. FB-Mamba achieves consistently high accuracy on both datasets, suggesting that the learned representations effectively encode disease-specific gait signatures such as stride-to-stride variability. The state space formulation is well suited for these tasks because neurodegenerative gait impairments manifest as subtle temporal fluctuations that require modeling of long-range dependencies. The adaptive fusion within the bidirectional Mamba blocks enables the model to dynamically balance information from preceding and subsequent strides. This mechanism may help capture the asymmetric gait patterns commonly observed in these conditions. Prospective clinical validation would be necessary to confirm real-world diagnostic utility.

For event-level detection, this category encompasses FoG detection on Daphnet and fall detection on MobiFall. Both tasks focus on identifying discrete clinical events that occur within continuous sensor streams. For FoG detection, the per-timestep sequence labeling formulation requires the model to identify precise onset and offset boundaries of freezing episodes. The bidirectional architecture is particularly relevant here because detecting FoG transitions benefits from observing both preceding movement deceleration and subsequent motor arrest. For fall detection, the window-level classification requires distinguishing abrupt fall events from activities of daily living based on characteristic acceleration and orientation changes. The Daphnet dataset exhibits significant class imbalance, with FoG episodes comprising approximately 8% of the total samples. Addressing such imbalance through specialized sampling or loss weighting strategies constitutes a distinct research direction that is beyond the scope of this work. Our focus is to validate the effectiveness and generalizability of the proposed bidirectional Mamba architecture across diverse task types rather than to optimize performance on individual datasets.

The results across all three task categories reveal several consistent findings. The ablation study (Table 4) confirms that all three architectural components make substantial contributions, and the adaptive fusion mechanism within the Mamba blocks outperforms simple concatenation. Unlike the fixed concatenation used in BiLSTMs, the adaptive fusion learns input-dependent weights through the selective state space mechanism, allowing the model to emphasize the more informative temporal direction at each timestep. This property is especially valuable for gait analysis, where the relative importance of past and future context varies across different phases

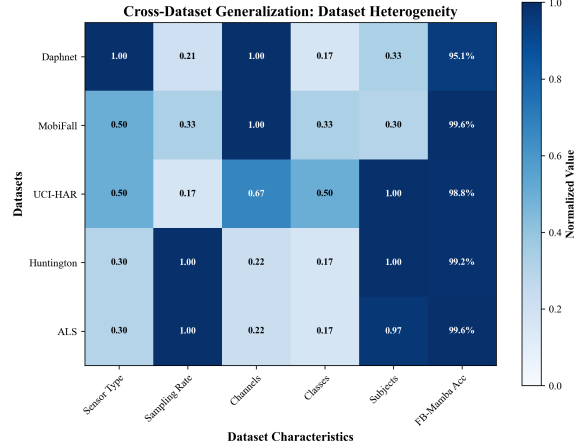


Figure 7: Cross-dataset generalization heatmap. Accuracy scores are shown with color intensity proportional to performance. Datasets are organized by task category. Darker colors indicate higher accuracy.

of the gait cycle. The long sequence experiments (Table 6) further demonstrate that FB-Mamba maintains favorable performance as sequence length increases compared to Transformer and LSTM, suggesting its suitability for continuous monitoring applications that require extended observation windows. The linear computational complexity (Table 3) enables efficient processing of such long sequences, which would be computationally prohibitive for Transformer-based approaches [21].

From a clinical perspective, the consistent performance across heterogeneous datasets suggests the potential of FB-Mamba as a general-purpose tool for wearable-based gait assessment. For neurodegenerative disease management, accurate and automated gait analysis can enable early detection of disease progression and provide objective measures for treatment evaluation. For FoG and fall detection, timely identification of these events can trigger real-time interventions such as auditory cueing or caregiver alerts, thereby reducing fall risk and improving patient safety [1, 8]. The linear computational complexity also makes FB-Mamba feasible for deployment in continuous monitoring scenarios where computational resources and battery life are constrained. As visualized in Fig. 7, the model achieves competitive performance across all five datasets without dataset-specific modifications, validating the generalizability of the architectural design across diverse sensor modalities, sampling rates, and clinical applications.

Several limitations should be acknowledged and suggest directions for future research. First, all datasets used in this study were collected in controlled laboratory or clinical environments, which may not fully represent the variability and noise encountered in free-living conditions; we therefore plan to validate FB-Mamba on data collected in free-living environments and to conduct prospective clinical trials with real-time model deployment. Second, prospective validation studies with independent expert annotations would further strengthen clinical applicability, and we aim to explore multi-modal sensor fusion and self-supervised pre-training on large-scale unlabeled gait data to improve model generalization [69]. Third, real-time deployment on edge devices [67] requires further optimization through knowledge distillation [70] and quantization [68] to meet the computational and power constraints of wearable platforms, and we intend to develop personalized adaptation strategies for individual patients and to investigate the integration of FB-Mamba into clinical workflow systems for long-term gait monitoring [71].

7 Conclusion

We presented FB-MAMBA, a Forward-Backward Mamba architecture for efficient bidirectional gait analysis. By combining parallel forward and backward state space models with an adaptive fusion mechanism, FB-MAMBA achieves competitive performance across five heterogeneous datasets while maintaining linear computational complexity. Our evaluation spans three task categories, namely human activity recognition (UCI-HAR, 98.83%), disease detection (Huntington 99.17%, ALS 99.56%), and event-level detection (Daphnet FoG 95.29%, MobiFall 99.58%). The unified architecture handles all tasks without task-specific modifications, achieving $1.22\times$ higher throughput than Transformer and favorable long-range dependency modeling with 79.1% accuracy at sequence length 2048 compared to 68.3% for Transformer. Despite substantial heterogeneity in sensor types, sampling rates, and application domains across datasets, FB-MAMBA consistently achieves competitive performance, validating its practical applicability as a general-purpose framework for wearable sensor-based gait analysis.

References

- [1] A. Mirelman, P. Bonato, R. Camicioli, T. D. Ellis, N. Giladi, J. L. Hamilton, C. J. Hass, J. M. Hausdorff, E. Pelosin, and Q. J. Almeida, “Gait impairments in parkinson’s disease,” *The Lancet Neurology*, vol. 18, no. 7, pp. 697–708, 2019.
- [2] J. M. Hausdorff, D. A. Rios, and H. K. Edelberg, “Gait variability and fall risk in community-living older adults: A 1-year prospective study,” *Archives of Physical Medicine and Rehabilitation*, vol. 82, no. 8, pp. 1050–1056, 2001.
- [3] H. Zhang, C. Wu, Y. Huang, R. Song, D. Zanotto, and S. K. Agrawal, “Fall risk prediction using instrumented footwear in institutionalized older adults,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 32, pp. 812–822, 2024.
- [4] J. Cai, Z. Guan, J. Wang, Z. Ding, Y. Li, R. Song, and H. Zhang, “Impact of gait parameters and their variability on fall risk assessment accuracy using wearable sensor,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 33, pp. 1996–2003, 2025.
- [5] S. Patel, H. Park, P. Bonato, L. Chan, and M. Rodgers, “A review of wearable sensors and systems with application in rehabilitation,” *Journal of Neuroengineering and Rehabilitation*, vol. 9, 2012.
- [6] W. Tao, T. Liu, R. Zheng, and H. Feng, “Gait analysis using wearable sensors,” *Sensors*, vol. 12, no. 2, pp. 2255–2283, 2012.
- [7] H. Zhang, D. Zanotto, and S. K. Agrawal, “Estimating cop trajectories and kinematic gait parameters in walking and running using instrumented insoles,” *IEEE Robotics and Automation Letters*, vol. 2, no. 4, pp. 2159–2165, 2017.
- [8] E. Rovini, C. Maremmani, and F. Cavallo, “How wearable sensors can support parkinson’s disease diagnosis and treatment: A systematic review,” *Frontiers in Neuroscience*, vol. 11, 2017.
- [9] S. Del Din, A. Godfrey, B. Galna, S. Lord, and L. Rochester, “Free-living gait characteristics in ageing and parkinson’s disease: Impact of environment and ambulatory bout length,” *Journal of Neuroengineering and Rehabilitation*, vol. 13, 2016.

- [10] J. G. Nutt, B. R. Bloem, N. Giladi, M. Hallett, F. B. Horak, and A. Nieuwboer, “Freezing of gait: Moving forward on a mysterious clinical phenomenon,” *The Lancet Neurology*, vol. 10, no. 8, pp. 734–744, 2011.
- [11] N. Giladi, T. A. Treves, E. S. Simon, H. Shabtai, Y. Orlov, B. Kandinov, D. Paleacu, and A. D. Korczyn, “Freezing of gait in patients with advanced parkinson’s disease,” *Journal of Neural Transmission*, vol. 108, pp. 53–61, 2001.
- [12] K. Chen, D. Zhang, L. Yao, B. Guo, Z. Yu, and Y. Liu, “Deep learning for sensor-based human activity recognition: Overview, challenges, and opportunities,” *ACM Computing Surveys*, vol. 54, no. 4, pp. 1–40, 2021.
- [13] J. Wang, Y. Chen, S. Hao, X. Peng, and L. Hu, “Deep learning for sensor-based activity recognition: A survey,” *Pattern Recognition Letters*, vol. 119, pp. 3–11, 2019.
- [14] N. M. Foumani, L. Miller, C. W. Tan, G. I. Webb, G. Forestier, and M. Salehi, “Deep learning for time series classification and extrinsic regression: A current survey,” *ACM Computing Surveys*, vol. 56, no. 9, pp. 1–45, 2024.
- [15] H. Zhang, Y. Guo, and D. Zanutto, “Accurate ambulatory gait analysis in walking and running using machine learning models,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 28, no. 1, pp. 191–202, 2020.
- [16] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [17] A. Graves and J. Schmidhuber, “Framewise phoneme classification with bidirectional lstm and other neural network architectures,” *Neural Networks*, vol. 18, no. 5-6, pp. 602–610, 2005.
- [18] M. Schuster and K. K. Paliwal, “Bidirectional recurrent neural networks,” *IEEE Transactions on Signal Processing*, vol. 45, no. 11, pp. 2673–2681, 1997.
- [19] H. Ismail Fawaz, G. Forestier, J. Weber, L. Idoumghar, and P.-A. Muller, “Deep learning for time series classification: A review,” *Data Mining and Knowledge Discovery*, vol. 33, no. 4, pp. 917–963, 2019.
- [20] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [21] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, “Informer: Beyond efficient transformer for long sequence time-series forecasting,” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 12, pp. 11 106–11 115, 2021.
- [22] H. Wu, J. Xu, J. Wang, and M. Long, “Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting,” *Advances in Neural Information Processing Systems*, vol. 34, pp. 22 419–22 430, 2021.
- [23] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, “Swin transformer: Hierarchical vision transformer using shifted windows,” *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 10 012–10 022, 2021.
- [24] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” *arXiv preprint arXiv:2312.00752*, 2023.

- [25] A. Gu, K. Goel, and C. Ré, “Efficiently modeling long sequences with structured state spaces,” *International Conference on Learning Representations (ICLR)*, 2022, outstanding Paper Honorable Mention.
- [26] A. Gu, A. Gupta, K. Goel, and C. Ré, “On the parameterization and initialization of diagonal state space models,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 35 971–35 983, 2022.
- [27] J. T. Smith, A. Warrington, and S. W. Linderman, “Simplified state space layers for sequence modeling,” *arXiv preprint arXiv:2208.04933*, 2023.
- [28] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, “Vision mamba: Efficient visual representation learning with bidirectional state space model,” *International Conference on Machine Learning (ICML)*, 2024.
- [29] Y. Liu, Y. Tian, Y. Zhao, H. Yu, L. Xie, Y. Wang, Q. Ye, J. Jiao, and Y. Liu, “Vmamba: Visual state space model,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2024, spotlight.
- [30] D. Y. Fu, T. Dao, K. K. Saab, A. W. Thomas, A. Rudra, and C. Ré, “Hungry hungry hippos: Towards language modeling with state space models,” *International Conference on Learning Representations (ICLR)*, 2023, spotlight.
- [31] M. Poli, S. Massaroli, E. Nguyen, D. Y. Fu, T. Dao, S. Baccus, Y. Bengio, S. Ermon, and C. Ré, “Hyena hierarchy: Towards larger convolutional language models,” *arXiv preprint arXiv:2302.10866*, 2023.
- [32] B. Peng, E. Alcaide, Q. Anthony, A. Albalak, S. Arcadinho, H. Cao, X. Cheng, M. Chung, M. Grella, K. He *et al.*, “Rwkv: Reinventing rnns for the transformer era,” *arXiv preprint arXiv:2305.13048*, 2023.
- [33] F. J. Ordóñez and D. Roggen, “Deep convolutional and lstm recurrent neural networks for multimodal wearable activity recognition,” *Sensors*, vol. 16, no. 1, p. 115, 2016.
- [34] N. Y. Hammerla, S. Halloran, and T. Ploetz, “Deep, convolutional, and recurrent models for human activity recognition using wearables,” *arXiv preprint arXiv:1604.08880*, 2016, extended version accepted at IJCAI.
- [35] S. Bai, J. Z. Kolter, and V. Koltun, “An empirical evaluation of generic convolutional and recurrent networks for sequence modeling,” *arXiv preprint arXiv:1803.01271*, 2018.
- [36] H. Ismail Fawaz, B. Lucas, G. Forestier, C. Pelletier, D. F. Schmidt, J. Weber, G. I. Webb, L. Idoumghar, P.-A. Muller, and F. Petitjean, “Inceptiontime: Finding alexnet for time series classification,” *Data Mining and Knowledge Discovery*, vol. 34, no. 6, pp. 1936–1962, 2020.
- [37] A. Dempster, F. Petitjean, and G. I. Webb, “Rocket: Exceptionally fast and accurate time series classification using random convolutional kernels,” *Data Mining and Knowledge Discovery*, vol. 34, no. 5, pp. 1454–1495, 2020.
- [38] J. Camps, A. Samà, M. Martín, D. Rodríguez-Martín, C. Pérez-López, J. M. M. Arostegui, J. Cabestany, A. Català, S. Alcaine, B. Mestre *et al.*, “Deep learning for freezing of gait detection in parkinson’s disease patients in their homes using a waist-worn inertial measurement unit,” *Knowledge-Based Systems*, vol. 139, pp. 119–131, 2018.

- [39] L. Sigcha, N. Costa, I. Pavón, S. Costa, P. Arezes, J. M. López, and G. De Arcas, “Deep learning approaches for detecting freezing of gait in parkinson’s disease patients through on-body acceleration sensors,” *Sensors*, vol. 20, no. 7, p. 1895, 2020.
- [40] E. E. Tripoliti, A. T. Tzallas, M. G. Tsipouras, G. Rigas, P. Bougia, M. Leontiou, S. Konitsiotis, M. Chondrogiorgi, S. Tsouli, and D. I. Fotiadis, “Automatic detection of freezing of gait events in patients with parkinson’s disease,” *Computer Methods and Programs in Biomedicine*, vol. 110, no. 1, pp. 12–26, 2013.
- [41] Z. Guan, J. Cai, J. Wang, Y. Li, R. Song, and D. Zanotto, “Accuracy and precision of wearable-derived gait parameters: How these affect the performance of models for fall prediction in the elderly,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 33, pp. 4255–4266, 2025.
- [42] G. Zerveas, S. Jayaraman, D. Patel, A. Bhamidipaty, and C. Eickhoff, “A transformer-based framework for multivariate time series representation learning,” *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, pp. 2114–2124, 2021.
- [43] H. Zhang, C. Wu, Y. Huang, X. Li, X. Ma, and R. Song, “Two-dimensional deep convolutional neural networks for estimating stride length and velocity in institutionalized older adults,” *IEEE Sensors Journal*, vol. 24, no. 17, pp. 27 766–27 776, 2024.
- [44] H. Zhang, T. T. H. Duong, A. K. Rao, P. Mazzone, S. K. Agrawal, and Y. Guo, “Transductive learning models for accurate ambulatory gait analysis in elderly residents of assisted living facilities,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 30, pp. 1155–1165, 2022.
- [45] J. Wang, Z. Guan, J. Cai, X. Li, C. Wu, and X. Ma, “Deep neural networks for gait cycle percentage prediction in frail older adults using a foot-mounted IMU,” in *2024 IEEE International Conference on Robotics and Biomimetics (ROBIO)*. IEEE, 2024, pp. 1–6.
- [46] T. Dao and A. Gu, “Transformers are ssms: Generalized models and efficient algorithms through structured state space duality,” *International Conference on Machine Learning (ICML)*, 2024.
- [47] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding,” *arXiv preprint arXiv:1810.04805*, 2019.
- [48] S. T. Moore, H. G. MacDougall, and W. G. Ondo, “Ambulatory monitoring of freezing of gait in parkinson’s disease,” *Journal of Neuroscience Methods*, vol. 167, no. 2, pp. 340–348, 2008.
- [49] S. Mazilu, M. Hardegger, Z. Zhu, D. Roggen, G. Tröster, M. Plotnik, and J. M. Hausdorff, “Online detection of freezing of gait with smartphones and machine learning techniques,” *Pervasive Computing Technologies for Healthcare*, pp. 123–130, 2012.
- [50] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, “Wearable assistant for parkinson’s disease patients with the freezing of gait symptom,” *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.
- [51] D. Anguita, A. Ghio, L. Oneto, X. Parra, and J. L. Reyes-Ortiz, “A public domain dataset for human activity recognition using smartphones,” vol. 3, p. 3, 2013.

- [52] J. M. Hausdorff, Y. Ashkenazy, C.-K. Peng, P. C. Ivanov, H. E. Stanley, and A. L. Goldberger, "Gait dynamics in parkinson's disease: common and distinct behavior among stride length, gait variability, and fractal-like scaling," *Chaos: An Interdisciplinary Journal of Nonlinear Science*, vol. 11, no. 2, pp. 392–397, 2001.
- [53] A. L. Goldberger, L. A. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley, "Physiobank, physiotoolkit, and physionet: components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.
- [54] G. Vavoulas, M. Pediaditis, E. G. Spanakis, and M. Tsiknakis, "The mobifall dataset: An initial evaluation of fall detection algorithms using smartphones," in *Proceedings of the 6th IEEE International Symposium on Monitoring and Surveillance Research (ISMSR): Healthcare-Bioinformatics*. Chania, Greece: IEEE, 2013, pp. 1–4.
- [55] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," *arXiv preprint arXiv:1711.05101*, 2017.
- [56] P. Prabhu, A. Karunakar, H. Anitha, and N. Pradhan, "Classification of gait signals into different neurodegenerative diseases using statistical analysis and recurrence quantification analysis," *Pattern Recognition Letters*, vol. 139, pp. 10–16, 2020.
- [57] F. Setiawan, A.-B. Liu, and C.-W. Lin, "Development of neuro-degenerative diseases' gait classification algorithm using convolutional neural network and wavelet coherence spectrogram of gait synchronization," *IEEE Access*, vol. 10, pp. 38 495–38 511, 2022.
- [58] N. Kleanthous, A. J. Hussain, W. Khan, and P. Liatsis, "A new machine learning based approach to predict freezing of gait," *Pattern Recognition Letters*, vol. 140, pp. 119–126, 2020.
- [59] K. Xia, J. Huang, and H. Wang, "Lstm-cnn architecture for human activity recognition," *IEEE Access*, vol. 8, pp. 56 855–56 866, 2020.
- [60] Ç. B. Erdaş, B. Ö. Olcay, and M. H. Özcanhan, "Classification of neurodegenerative diseases using gait dynamics via the ground reaction force sensor," *Multimedia Tools and Applications*, vol. 82, pp. 36 317–36 337, 2023.
- [61] A.-G. Andrei, A.-M. Tautan, and B. Ionescu, "Freezing of gait detection in parkinson's disease: comparison of deep learning frameworks," in *2024 IEEE International Symposium on Medical Measurements and Applications (MeMeA)*. IEEE, 2024, pp. 1–6.
- [62] S. Wan, L. Qi, X. Xu, C. Tong, and Z. Gu, "Deep learning models for real-time human activity recognition with smartphones," *Mobile Networks and Applications*, vol. 25, no. 2, pp. 743–755, 2020.
- [63] S. Yu, Y. Chai, H. Chen, R. A. Brown, S. J. Sherman, and J. F. Nunamaker Jr., "Fall detection with wearable sensors: A hierarchical attention-based convolutional neural network approach," *Journal of Management Information Systems*, vol. 38, no. 4, pp. 1095–1121, 2021.
- [64] F. Alfayez and S. B. Khan, "Iot-blockchain empowered trinet: Optimized fall detection system for elderly safety," *Frontiers in Bioengineering and Biotechnology*, vol. 11, p. 1257676, 2023.
- [65] J. He, Z. Zhang, X. Wang, and S. Yang, "A low power fall sensing technology based on fd-cnn," *IEEE Sensors Journal*, vol. 19, no. 13, pp. 5110–5118, 2019.

- [66] J. Zhang, B. Yang, L. Tang, L. Zhou, Z. Jiang, Z. Liu, and Y. Liu, “An efficient FoG-M3 method for self-adaptive labeling and predicting freezing of gait,” *IEEE Journal of Biomedical and Health Informatics*, vol. 30, no. 1, pp. 184–195, 2026.
- [67] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, “Mobilenets: Efficient convolutional neural networks for mobile vision applications,” *arXiv preprint arXiv:1704.04861*, 2017.
- [68] S. Han, H. Mao, and W. J. Dally, “Deep compression: Compressing deep neural networks with pruning, trained quantization and huffman coding,” *arXiv preprint arXiv:1510.00149*, 2015.
- [69] J. Wang, Z. Guan, T. Liang, Z. Ding, X. Ma, and Y. Li, “Multi-task learning for gait phase and gait cycle percentage prediction with wearable sensors in frail older adults,” *IEEE Journal of Biomedical and Health Informatics*, pp. 1–13, 2025, early Access.
- [70] G. Hinton, O. Vinyals, and J. Dean, “Distilling the knowledge in a neural network,” *arXiv preprint arXiv:1503.02531*, 2015.
- [71] H. Zhang, S. Li, Q. Zhao, A. K. Rao, Y. Guo, and D. Zanutto, “Reinforcement learning-based adaptive biofeedback engine for overground walking speed training,” *IEEE Robotics and Automation Letters*, vol. 7, no. 3, pp. 8487–8494, 2022.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [FBMambaOverleaf.zip](#)